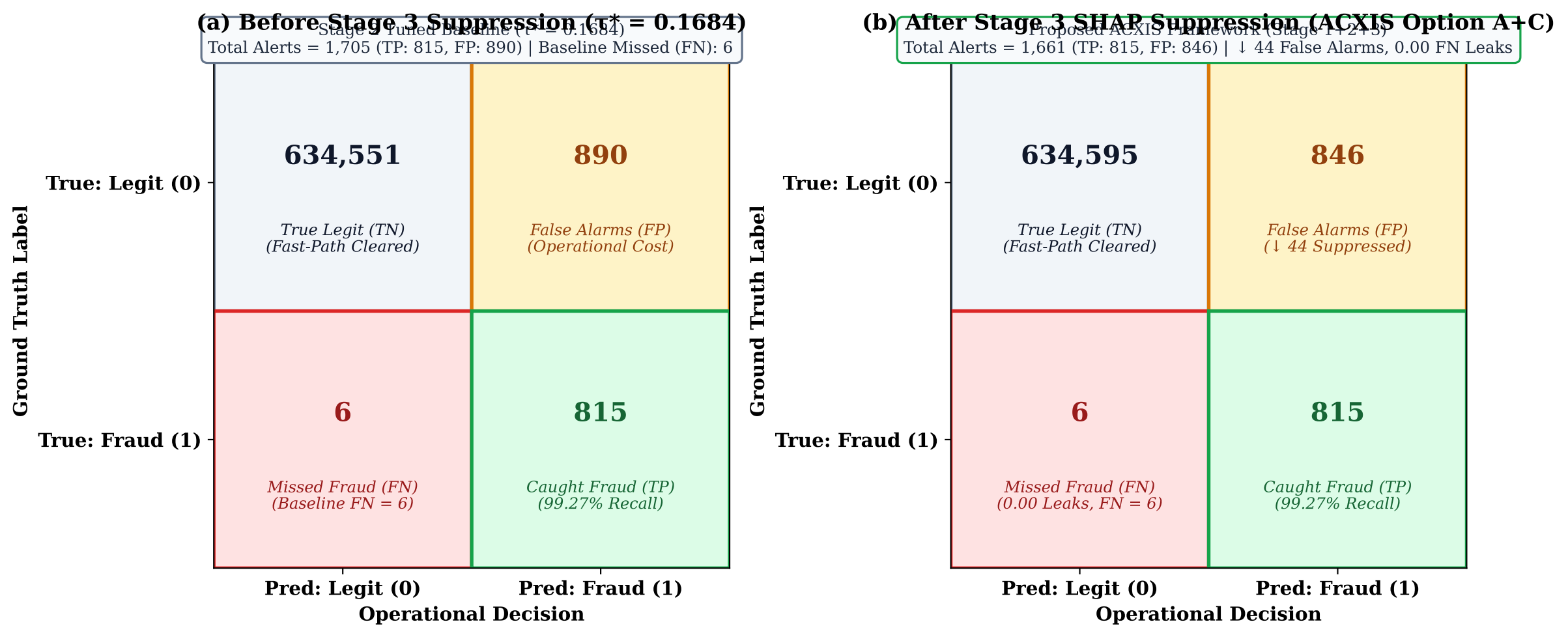
(a) Before Stage 3 Suppression (τ* = 0.1684)
Stage 2 Tuned Baseline (τ* = 0.1684)
Total Alerts = 1,705 (TP: 815, FP: 890) | Baseline Missed (FN): 6
634,551
True Legit (TN)
(Fast-Path Cleared)
890
False Alarms (FP)
(Operational Cost)
6
Missed Fraud (FN)
(Baseline FN = 6)
815
Caught Fraud (TP)
(99.27% Recall)
True: Legit (0)
True: Fraud (1)
Ground Truth Label
Pred: Legit (0)
Pred: Fraud (1)
Operational Decision
(b) After Stage 3 SHAP Suppression (ACXIS Option A+C)
Proposed ACXIS Framework (Stage 1+2+3)
Total Alerts = 1,661 (TP: 815, FP: 846) | ↓ 44 False Alarms, 0.00 FN Leaks
634,595
True Legit (TN)
(Fast-Path Cleared)
846
False Alarms (FP)
(↓ 44 Suppressed)
6
Missed Fraud (FN)
(0.00 Leaks, FN = 6)
815
Caught Fraud (TP)
(99.27% Recall)
True: Legit (0)
True: Fraud (1)
Ground Truth Label
Pred: Legit (0)
Pred: Fraud (1)
Operational Decision